Classification

Random Forest:

	Predicted 0	Predicted 1
Actual 0	78 TN	7 FP
Actual 1	6 FN	43 TP

1. How many customers were correctly predicted as Not Purchased?
   TN – 78
2. How many customers were correctly predicted as Purchased?
   TP -- 43
3. What is the overall accuracy of the model?
   78+43/(78+7+6+43) – 90%
4. When the model predicts "Purchased", how often is it correct?
   43/43+7 – 86%
5. Out of all customers who purchased, how many did the model identify?
   43/43+6 – 88%
6. What is the F1-score for Purchased?
   2*0.86*0.88/0.86+0.88 – 87%
7. What is the Macro Average Precision?
   0.93+0.86/2 – 89%
8. What is the Macro Average F1-score?

   0.92+0.87/2 – 90%

9. What is the Weighted Average F1-score?

   0.92*85 + 0.87 *49 /134 – 90%

Decision Tree:

	Predicted 0	**Predicted 1**
Actual 0	TN = 76	FP = 9
Actual 1	FN = 8	TP = 41

1. What is the overall accuracy of the model?
   (41 76) / (41 + 76 + 9 + 8) – 87%
2. When the model predicts "Purchased", how often is it correct?

   41 / (41 + 9) – 82%
3. What is the Precision for Not Purchased?
   76 / (76 + 8) – 90%
4. What is the Recall for Purchased?

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41 / (41 + 8) – 84%

5. What is the Recall for Not Purchased?
   76 / (76 + 9) – 89%
6. What is the F1-Score for Purchased?
   2 × (0.82 × 0.8367) / (0.82 + 0.8367) – 83%
7. What is the F1-Score for Not Purchased?
   2 × (0.9048 × 0.8941) / (0.9048 + 0.8941) – 90%
8. What is the Macro Average F1-Score?
   (0.8994 + 0.8284) / 2 - 86%
9. What is the Weighted Average F1-Score?
   (0.8994 × 85 + 0.8284 × 49) / 134 – 87%

Support Vector Machine:

	Predicted 0	**Predicted 1**
Actual 0	TN = 82	FP = 3
Actual 1	FN = 26	TP = 23

1. What is the overall accuracy of the model?

   (23 + 82) / (23 + 82 + 3 + 26) – 78%

2. When the model predicts "Purchased", how often is it correct?

   23 / (23 + 3) – 88%

3. What is the Precision for Not Purchased?

   82 / (82 + 26) -76%

4. What is the Recall for Purchased?

   23 / (23 + 26) – 47%

5. What is the Recall for Not Purchased?

   82 / (82 + 3) – 96%

6. What is the F1-Score for Purchased?

   2 × (0.88 × 0.47) / (0.88 + 0.47) – 61%

7. What is the F1-Score for Not Purchased (Class 0)?

   2 × (0.76 × 0.96) / (0.76 + 0.96) – 85%

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8. What is the Macro Average F1-Score?

(0.85 + 0.61) / 2 - 73%

9. What is the Weighted Average F1-Score?

(0.85 × 85 + 0.61 × 49) / 134 – 76%